List of Symbols

Δf_{osc}	Bandwidth of the oscillation frequency
ΔT	Time step size in dynamic calculations
$\hat{\omega}_e$	Electrical angular velocity
$\hat{\omega}_{\text{mech}}$	Mechanical angular velocity
$\hat{i}_{\text{dc}}$	AC component of the DC current
$\hat{v}_{\text{bus}}$	AC component of the DC bus voltage
λ	Flux linkage of the motor
ω_x	Frequency of the zero in the transfer function
ω_{bw}	Bandwidth of the system
ω_{res}	Resonance frequency of the filter
ω_{sys}	System frequency
a_0, a_1, a_2	Numerator coefficients of the transfer function
b_0, b_1, b_2	Denominator coefficients of the transfer function
C_{filter}	Filter capacitance
D_d	Direct axis duty cycle
D_q	Quadrature axis duty cycle
f_{osc}	Oscillation frequency of the system
G_c	Controller transfer function gain
I_{in}	Input current of the voltage sensor
I_{max}	Maximum permissible input current
I_{peak}	Peak current through the measurement resistor
I_d	Direct axis current
I_d^{ref}	Direct axis reference current
I_q	Quadrature axis current
I_q^{ref}	Quadrature axis reference current

I_{bus}	Current flowing through the DC bus
K	Controller gain constant
L_q	Quadrature axis inductance
$M(s)$	Mechanical transfer function of the motor
P_{cpl}	Power of the constant power load
Q_R	Resonance quality factor
r	Resistance of the motor windings
R_1	Series resistor for voltage sensing
R_M	Measurement resistor in the circuit
R_{filter}	Filter resistance
R_{TVS}	TVS diode clamping resistance
S_d	Sensitivity function with damping compensation
S_n	Sensitivity function of the system
$T(s)$	Transfer function of the feedback system
T_{op}	Open-loop transfer function
T_{PLL}	Adaptive phase-locked loop tracking frequency
V_{ADC}	Voltage output of the ADC channel
V_{max}	Maximum permissible input voltage
V_{out}	Output voltage of the voltage sensor
V_d	Direct axis voltage
V_d^{ref}	Direct axis reference voltage
V_q	Quadrature axis voltage
V_q^{ref}	Quadrature axis reference voltage
V_{bus}	DC bus voltage
Z_d	Input impedance of the system with damping
Z_n	Equivalent impedance of the system
$Z_{\text{in}}(s)$	Input impedance of the system
$Z_{\text{out}}(s)$	Output impedance of the system
$Z_{\text{VIR}}(s)$	Virtual impedance in the system